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 Studierichting: Informatica
 Oefeningenreeks 3F nr. 2.5

Bepaal de schuine asymptoot door een euclidische deling te maken:

$$f(x) = \frac{\sqrt{2}x^2 + \sqrt{3}x - \sqrt{2}}{x + \sqrt{6}}$$

$$\begin{array}{r|l}
 \begin{array}{rrr}
 +\sqrt{2}x^2 & +\sqrt{3}x & -\sqrt{2} \\
 - & +\sqrt{2}x^2 & +2\sqrt{3}x \\
 & -\sqrt{3}x & -\sqrt{2} \\
 & - & -\sqrt{3}x & -3\sqrt{2} \\
 & & +2\sqrt{2}
 \end{array}
 & \begin{array}{l}
 x + \sqrt{6} \\
 \hline
 \sqrt{2}x - \sqrt{3}
 \end{array}
 \end{array}$$

Schuine asymptoot:

$$A : y = \sqrt{2}x - \sqrt{3}$$

$$f(x) - A = \frac{2\sqrt{2}}{x + \sqrt{6}}$$

$$\begin{array}{r|l}
 x & -\sqrt{6} \\
 \hline
 f(x) - A & - \quad +
 \end{array}$$

Dus:

$$\lim_{x \rightarrow +\infty} (f(x) - A) = 0^+ \Rightarrow x \rightarrow +\infty : f > A$$

$$\lim_{x \rightarrow -\infty} (f(x) - A) = 0^- \Rightarrow x \rightarrow -\infty : f < A$$

References